

Robust CIFAR-10 Classification Under Symmetric and Asymmetric Label Noise

CS725 Foundations of Machine Learning Project

Abstract

Deep neural networks can achieve high accuracy on image classification benchmarks, but their performance often degrades when training labels are corrupted. This project studies robust image classification on CIFAR-10 under synthetic symmetric and asymmetric label noise. A custom ResNet-18-based model was implemented using labels represented in isometric log-ratio (ILR) space, an uncertainty-aware multivariate Gaussian loss, and exponential moving average (EMA) based label correction. The method was compared with a standard ResNet-18 cross-entropy baseline across multiple noise rates. The custom method achieved consistent improvements in most moderate and high noise settings, with a maximum test accuracy gain of 17.38 percentage points under symmetric noise and 16.00 percentage points under asymmetric noise.

1 Problem Statement

The objective of this project is to evaluate whether a classifier trained with uncertainty-aware label modeling can be more robust than a conventional cross-entropy classifier when the training labels are noisy. The setting is CIFAR-10 image classification with controlled synthetic label corruption. The project compares:

- a baseline ResNet-18 trained using cross-entropy loss, and
- a custom ResNet-18-based model trained in transformed label space using a multivariate Gaussian objective and EMA-based correction.

The central question is: how does the custom model perform relative to the baseline as label noise increases?

2 Introduction and Motivation

Supervised deep learning assumes that training labels are reliable. In practice, labels collected from weak supervision, web scraping, crowdsourcing, or imperfect annotation pipelines can contain substantial noise. A model trained with ordinary cross-entropy may memorize these corrupted labels, reducing generalization on clean test data.

CIFAR-10 is a useful benchmark for studying this issue because it has a clean test set and well-defined image classes. By injecting synthetic noise into the training labels, one can measure how robust each training method is as the signal-to-noise ratio decreases.

This project focuses on two realistic forms of label corruption:

- **Symmetric noise:** labels are randomly changed to another class.

- **Asymmetric noise:** labels are changed to semantically related classes, such as cat to dog or automobile to truck.

Symmetric noise tests general robustness to random corruption. Asymmetric noise is more structured and resembles mistakes that may occur during human or weak-label annotation.

3 Literature Review

Research on noisy-label learning has shown that deep networks can eventually memorize corrupted labels, especially when trained for many epochs [7]. Reed et al. introduced a bootstrapping approach that mixes ground-truth labels with model predictions to reduce the influence of noisy targets [6]. Patrini et al. proposed loss correction using estimated label transition matrices [5]. Zhang and Sabuncu introduced generalized cross-entropy as a loss that interpolates between mean absolute error and cross-entropy for improved robustness [8]. Han et al. proposed co-teaching, where two networks exchange small-loss examples during training [3]. DivideMix later treated noisy-label learning as a semi-supervised learning problem by separating likely clean and noisy samples [4].

The label representation in this project is also connected to compositional data analysis. Class-probability vectors lie on a simplex because their components are non-negative and sum to one. Egozcue et al. introduced the isometric log-ratio transform as a way to represent simplex-valued compositions in Euclidean space [2]. Andrews et al. used ILR-style transformations for probabilistic labels, showing that probability-vector targets can be handled with standard Euclidean modeling after a suitable transform [1]. The present project combines this geometric label representation with a neural classifier and a Gaussian loss in transformed label space.

4 Methodology

4.1 Data Preparation

The dataset used is CIFAR-10, consisting of 60,000 color images across 10 classes. The 50,000 training images were split into 80% training and 20% validation subsets. The 10,000-image test set was kept clean and used only for final evaluation.

The following preprocessing was applied:

- random crop with padding for training images,
- conversion to tensor,
- CIFAR-10 channel-wise normalization.

4.2 Noise Injection

For symmetric noise, each training label is independently flipped with probability η to one of the other nine classes.

For asymmetric noise, each selected label is changed according to a fixed semantic mapping:

Original Class	Noisy Class
airplane	bird
automobile	truck
bird	airplane
cat	dog
deer	horse
dog	cat
frog	deer
horse	deer
ship	automobile
truck	automobile

4.3 Baseline Model

The baseline is a ResNet-18 adapted for CIFAR-10. The first convolution was changed to a 3×3 stride-1 convolution, the initial max-pooling layer was removed, and the final fully connected layer was replaced with a 10-class classifier. The model was trained using Adam and cross-entropy loss.

4.4 Custom Model

The custom model uses the same ResNet-18 feature extractor but replaces the final classifier with two heads:

- a mean head $\mu \in \mathbb{R}^9$,
- a covariance-factor head $r \in \mathbb{R}^9$.

The 10-class label vector is smoothed and then transformed from the probability simplex to a 9-dimensional Euclidean representation using the ILR transform. If p is the smoothed class-probability vector and H is the Helmert sub-matrix, the transformed target is:

$$z = H \left(\log p - \frac{1}{K} \sum_{j=1}^K \log p_j \right), \quad (1)$$

where $K = 10$ for CIFAR-10.

The model predicts the mean of a Gaussian distribution in this ILR space. The covariance is parameterized as:

$$\Sigma = I + rr^T. \quad (2)$$

The loss is the negative log-likelihood form:

$$\mathcal{L} = \frac{1}{2}(z - \mu)^T \Sigma^{-1}(z - \mu) + \frac{1}{2} \log |\Sigma|. \quad (3)$$

This objective encourages the model to learn both class prediction and uncertainty structure in the transformed label space.

4.5 EMA-Based Label Correction

After the initial training phase, an exponential moving average copy of the model is used to estimate a more stable correction signal. The correction is introduced after epoch 20, allowing the model to first learn basic visual structure before relying on EMA predictions. This is intended to reduce overfitting to corrupted labels in later epochs.

5 Experimental Setup

All models were trained for 60 epochs per noise rate with batch size 128 and Adam optimizer using learning rate 10^{-3} . Evaluation used macro F1 score and clean test-set accuracy.

The symmetric noise experiments used noise rates:

0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9.

The asymmetric noise experiments used noise rates:

0.1, 0.2, 0.3, 0.4, 0.5, 0.6.

6 Results

6.1 Symmetric Noise

Table 1: Test performance under symmetric label noise. Accuracy gain is custom accuracy minus baseline accuracy.

Noise Rate	Custom Acc.	Baseline Acc.	Custom F1	Gain
0.1	85.07%	81.74%	0.8513	+3.33
0.2	82.83%	75.75%	0.8267	+7.08
0.3	78.78%	65.92%	0.7873	+12.86
0.4	70.71%	59.10%	0.7042	+11.61
0.5	62.26%	45.08%	0.6188	+17.18
0.6	55.65%	38.27%	0.5564	+17.38
0.7	39.82%	27.35%	0.3967	+12.47
0.8	29.22%	16.50%	0.2824	+12.72
0.9	9.40%	10.59%	0.0736	-1.19

The custom model outperformed the baseline in 8 of 9 symmetric-noise settings. The strongest improvement was at 60% noise, where the custom model achieved 55.65% test accuracy compared with 38.27% for the baseline.

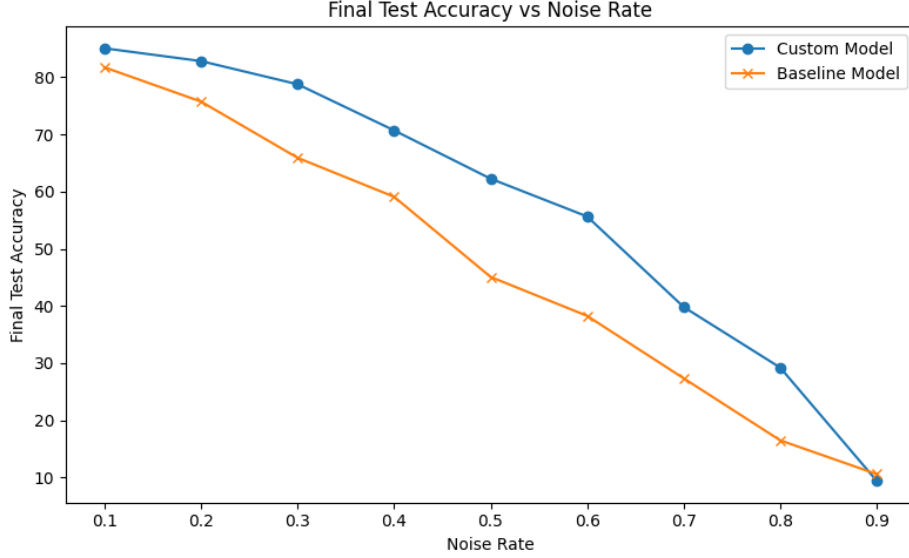


Figure 1: Final test accuracy comparison under symmetric noise.

6.2 Asymmetric Noise

Table 2: Test performance under asymmetric label noise. Accuracy gain is custom accuracy minus baseline accuracy.

Noise Rate	Custom Acc.	Baseline Acc.	Custom F1	Gain
0.1	83.26%	81.42%	0.8326	+1.84
0.2	82.02%	73.06%	0.8193	+8.96
0.3	75.69%	59.69%	0.7562	+16.00
0.4	72.41%	57.77%	0.7220	+14.64
0.5	51.96%	41.95%	0.5079	+10.01
0.6	23.23%	35.13%	0.2204	-11.90

The custom model outperformed the baseline in 5 of 6 asymmetric-noise settings. The best improvement was at 30% noise, where the custom model gained 16.00 percentage points over the baseline. At 60% asymmetric noise, however, the baseline performed better, suggesting that the correction mechanism became unreliable when the structured label corruption was too strong.

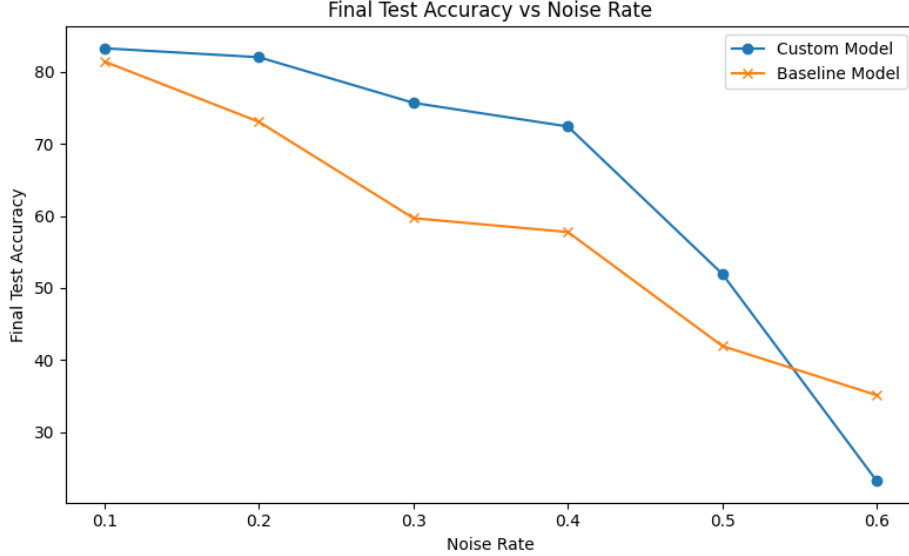


Figure 2: Final test accuracy comparison under asymmetric noise.

6.3 Overall Comparison

Across symmetric noise, the average gain of the custom model over the baseline was 10.38 percentage points. Across asymmetric noise, the average gain was 6.59 percentage points. The results show that the proposed method is most effective when labels are substantially noisy but not completely dominated by corruption.

7 Discussion

The results support the hypothesis that explicitly modeling label uncertainty can improve robustness to noisy labels. Cross-entropy treats the observed label as the single correct target, which can make the baseline sensitive to corrupted labels. The custom method instead maps smoothed labels into ILR space and optimizes a Gaussian objective with a learned covariance structure. This gives the model a richer way to represent uncertainty around labels.

The EMA correction also follows an important intuition: early training should learn broad visual features, while later training should avoid memorizing noisy labels. By introducing correction after epoch 20, the method attempts to use a more stable model estimate once the network has learned useful structure.

The failure cases are also informative. At 90% symmetric noise, almost all label information is corrupted, so neither method has enough reliable supervision. At 60% asymmetric noise, the corruption is structured and systematic, which can mislead correction-based methods because the wrong labels are not random outliers but consistent class-dependent targets.

8 What We Learned

- Robust loss design matters most in moderate-to-high noise regimes, where the baseline begins to overfit noisy labels.
- Representing class labels as simplex-valued compositions and transforming them with ILR provides a principled way to apply Euclidean probabilistic losses.

- EMA-based correction can improve robustness, but it depends on the model having learned a reliable signal before correction begins.
- Extremely high noise remains difficult, especially when corruption is systematic rather than random.
- Evaluation across both symmetric and asymmetric noise is important because the two settings stress different failure modes.

9 Conclusion

This project implemented and evaluated a noise-robust CIFAR-10 classifier based on ILR-transformed labels, multivariate Gaussian loss, and EMA-based label correction. Compared with a standard ResNet-18 cross-entropy baseline, the custom model achieved stronger performance across most noise levels. The largest improvements were 17.38 percentage points under symmetric noise and 16.00 percentage points under asymmetric noise.

The project shows that geometric label representations and uncertainty-aware objectives are promising tools for robust supervised learning. At the same time, the experiments show that robustness has limits: when label corruption becomes too extreme or too systematically biased, correction mechanisms can fail. Future work could add fixed random seeds, stronger data augmentation, confidence-based sample selection, and comparisons against established methods such as Co-teaching and DivideMix.

References

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